AGRI*INTEL*

TEAM : INTELLI*CORE*

THIS REPORT PRESENTS A STRUCTURED EXPLORATORY ANALYSIS OF 43,173 FARMER RECORDS ACROSS 105 FEATURES, AND OUR KEY FINDINGS ALONG WITH DETECTION OF PARAMETERS FOR INCOME GROWTH AND SEASONAL VOLTALITY MODELLING.

PRESENTED BY -

SHIVANSHI *TYAGI*

VITIKA *MADHWANI*

SHRAVYA *CHOWDHARY G*

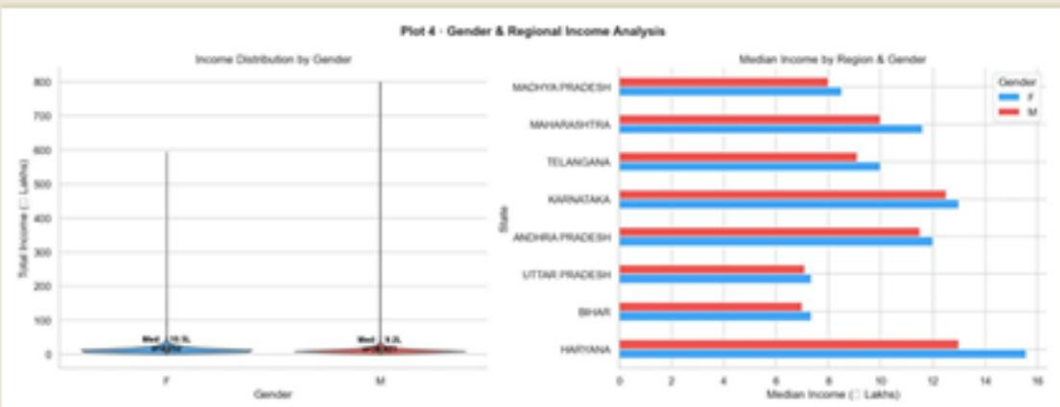
SAHAJ *ANEJA*

OUR *GOAL*

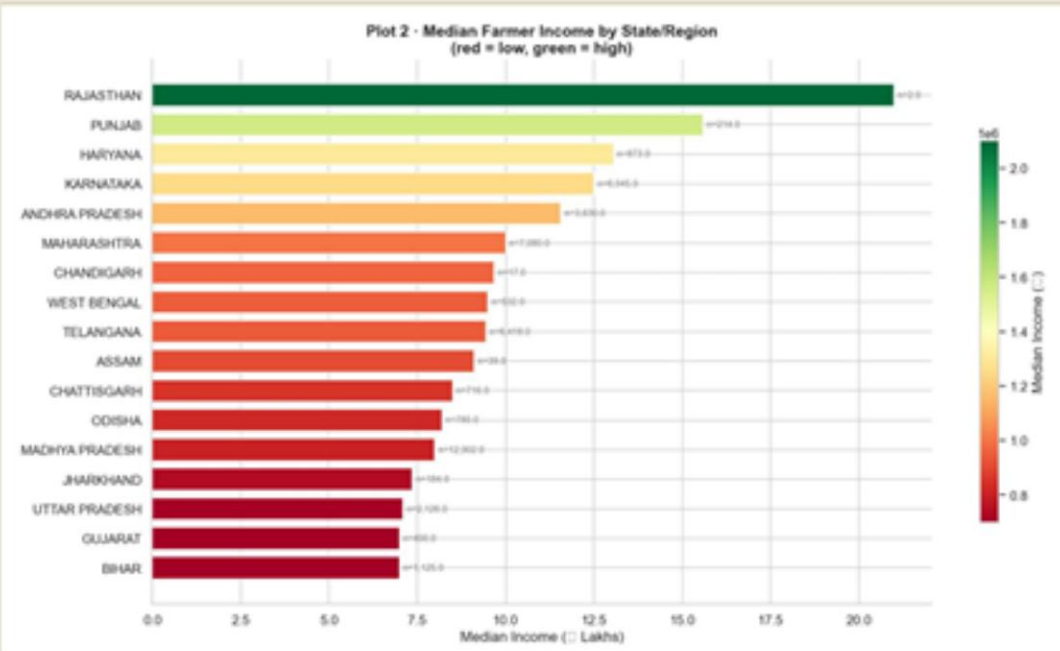
Our objective is to engineer a robust machine learning framework that deciphers the 'web of interacting variables'—from soil health and multi-year weather cycles to market accessibility and village-level socio-economic indicators. By linking these diverse signals to the Total Income target variable, the dataset serves as a powerful foundation for building predictive models that assess creditworthiness in the absence of traditional financial histories. Beyond income prediction, our mission encompasses a prescriptive strategy aimed at identifying actionable levers for economic growth. By isolating the impact of these variables, we can provide stakeholders with a roadmap of localized improvements.

Income distribution insights from plots

GENDER & REGIONAL INCOME ANALYSIS



MEDIAN INCOME BY STATE / REGION



On the gender front, the data challenges a common assumption — female farmers (Med: ₹10.5L) actually earn higher median incomes than male farmers (Med: ₹9.2L), though female participation is far smaller (n=4,252 vs n=38,921). This suggests that women who do engage in farming may operate in relatively higher-income contexts, or own larger landholdings on average.

Female farmers represent only about 10% of the sample, suggesting farming remains a male-dominated occupation in India. The income gap likely reflects a *selection effect* — women entering farming may do so under more favourable conditions, such as in wealthier states like Haryana, where the female income advantage is most pronounced.

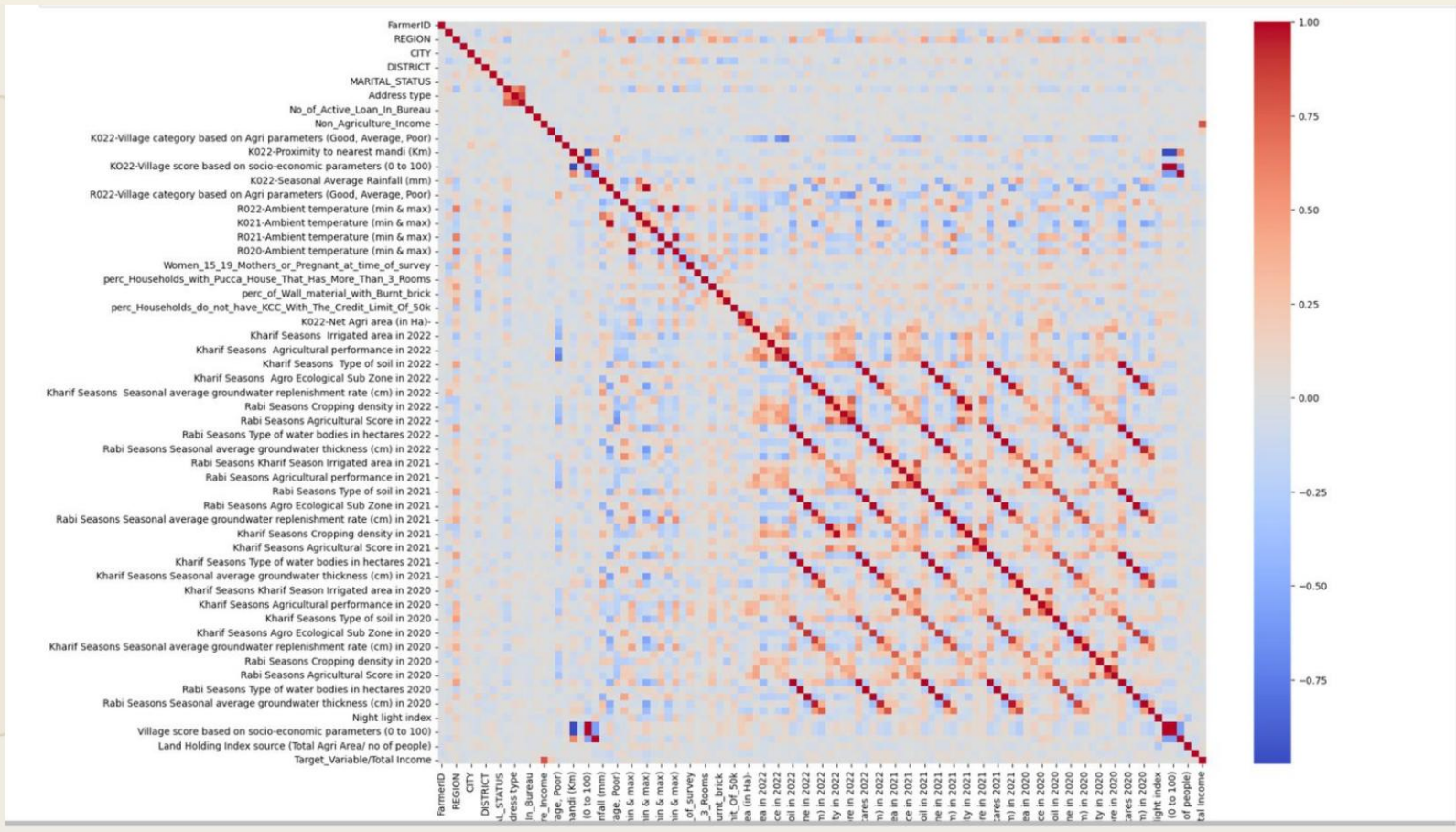
GENDER DISPARITY AT A GLANCE

FEMALE · MEDIAN	MALE · MEDIAN
₹10.5L	₹9.2L
n = 4,252 (≈10%)	n = 38,921 (≈90%)

State-level income disparities are significant, with high-performing states like **Haryana, Karnataka, and Andhra Pradesh** recording median incomes nearly twice that of lagging states like **Bihar, Uttar Pradesh, and Gujarat** — pointing to structural differences in irrigation, market access, and agricultural infrastructure as key drivers of regional inequality.

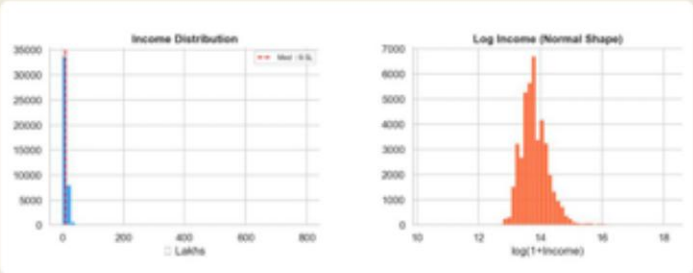
HEATMAP INSIGHTS

Kharif and Rabi variables from the same year form strong correlated clusters (dark-red diagonal blocks). **Non-Agriculture Income** shows a dominant positive correlation with Total Income — far above all other features. Groundwater and temperature variables show mild negative correlations. The block structure across years indicates significant **multicollinearity**, suggesting dimensionality reduction may be beneficial.



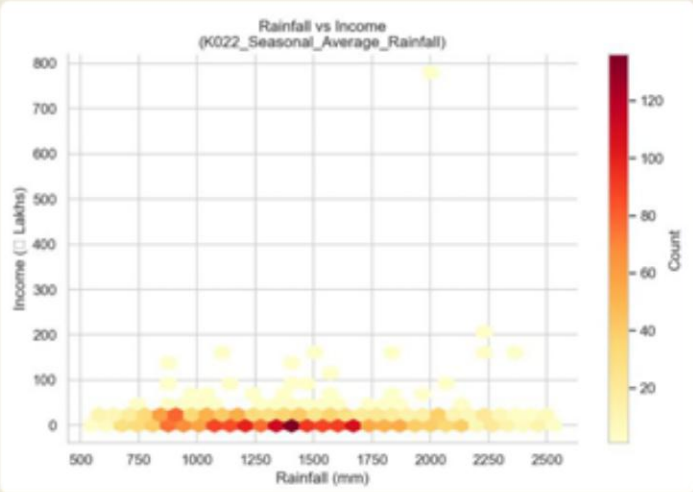
WHAT REALLY DRIVES *A FARMER'S INCOME?*

Meet the Data



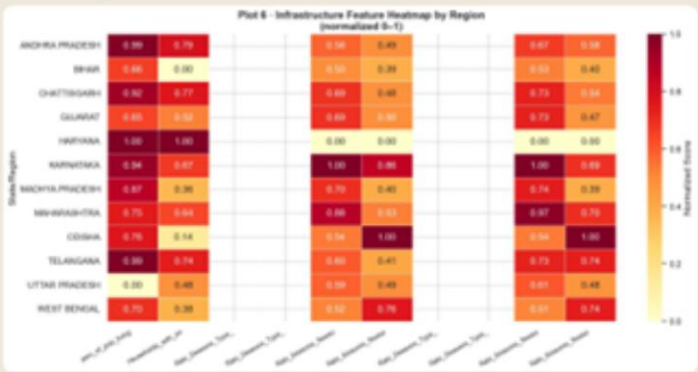
Income is heavily right-skewed — most farmers earn ₹5–20L with extreme outliers. A log transform reveals a near-normal shape, confirming *inequality is real but tractable.*

The Land Beneath Their Feet



Rainfall alone doesn't predict income — farmers cluster at the bottom regardless. *It's not nature alone — it's what the state built around it.*

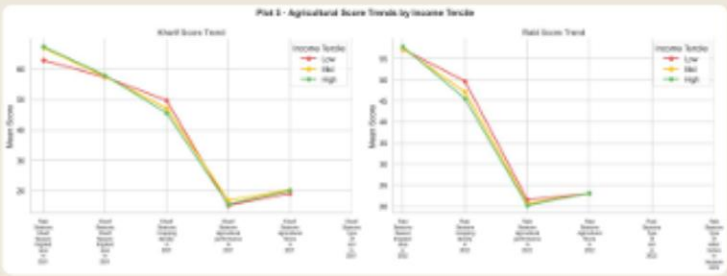
Infrastructure Tells the Real Story



States with better household connectivity & irrigation (Karnataka, Maharashtra) consistently outperform. *Infrastructure, not rainfall, separates regions.*

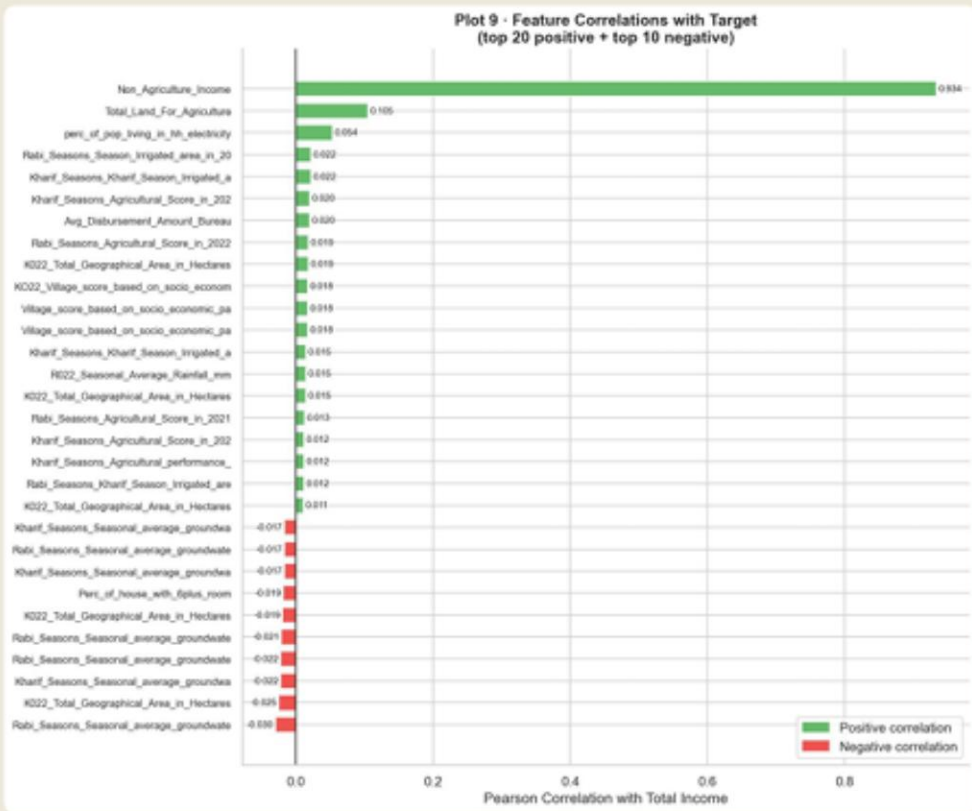
Does Farming Itself Matter?

Kharif & Rabi agricultural scores move almost identically across low, mid, and high income terciles. The gap between them is *surprisingly small* — farming quality alone doesn't separate rich from poor farmers. All income groups respond to the same seasonal pressures.

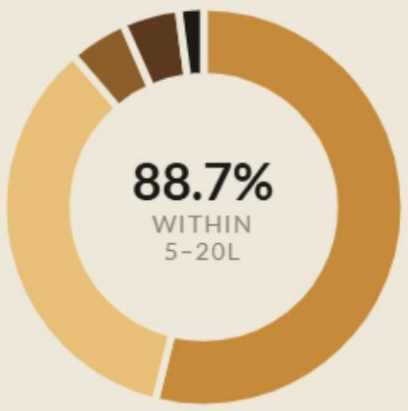


Then What Does?

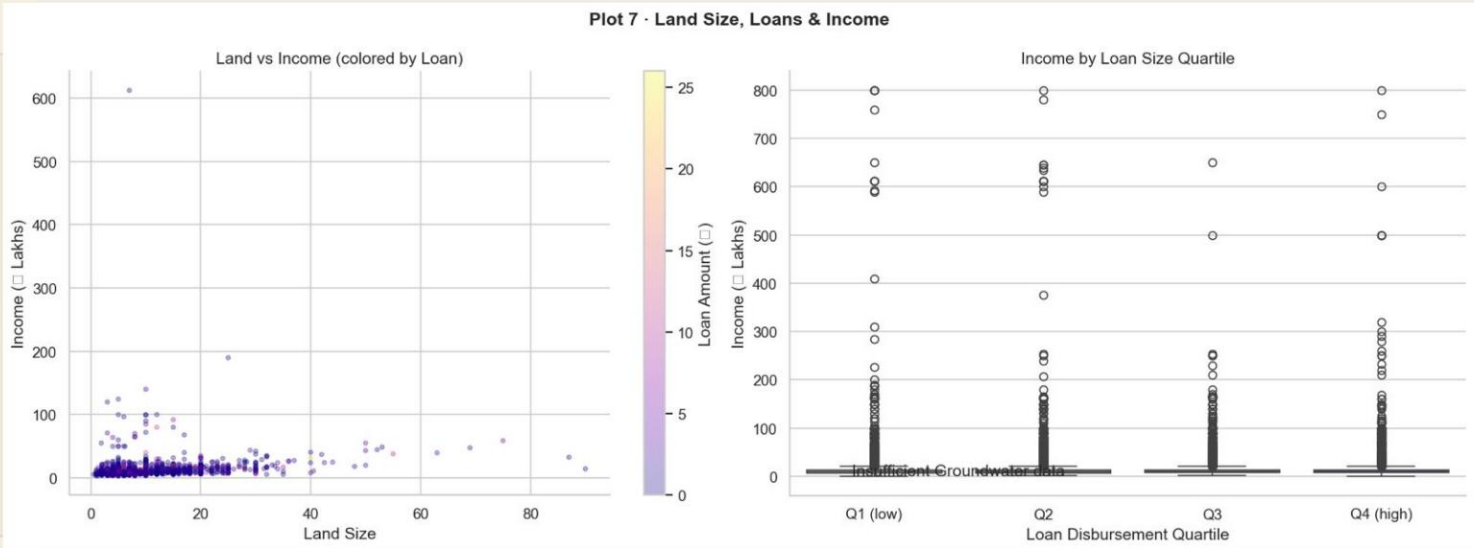
Feature correlations reveal the real driver — *non-farm income dominates everything else by a massive margin.* Land size is a distant second. Groundwater is negatively correlated.



INCOME BAND BREAKDOWN



- 53.9% ₹5–10L
- 34.8% ₹10–20L
- 4.7% ₹20–50L
- 4.6% <₹5L
- 2.0% >₹50L

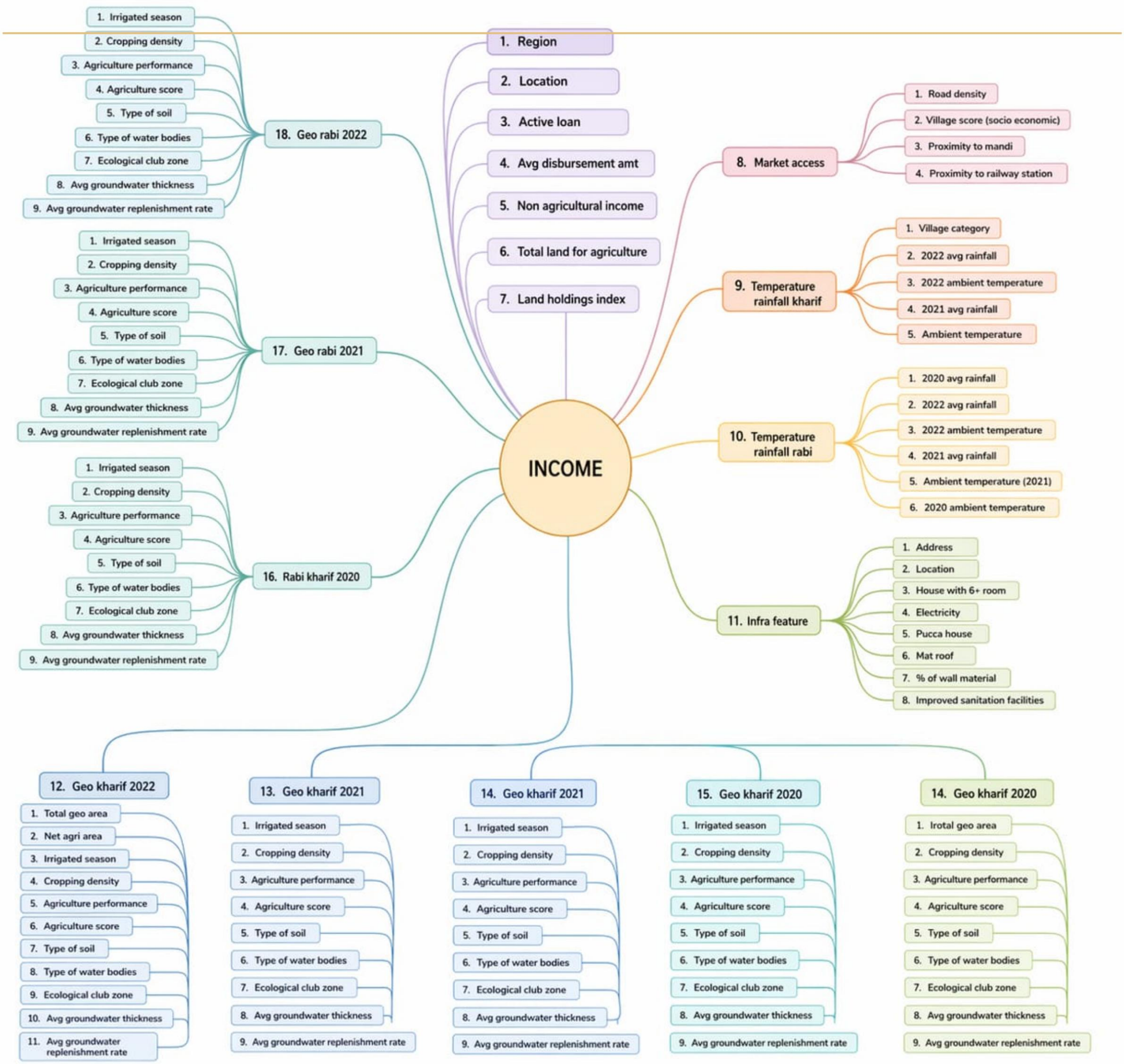


The vast majority of farmers **(~88.7%)** fall within the ₹5–20L income range, with over half concentrated in the ₹5–10L band — underscoring the urgency of targeted income uplift programs.

LAND SIZE, LOANS & INCOME · INSIGHTS

Land size shows a **weak positive relationship** with income — most farmers cluster below ₹50L regardless of holding size. Higher loan disbursements (Q3–Q4) do not consistently translate to higher incomes, suggesting **credit alone is insufficient**. Income outliers appear uniformly across all loan quartiles, indicating structural factors like market access and irrigation play a far stronger role than loan size.

FEATURE *ENGINEERING*



MODEL SELECTION

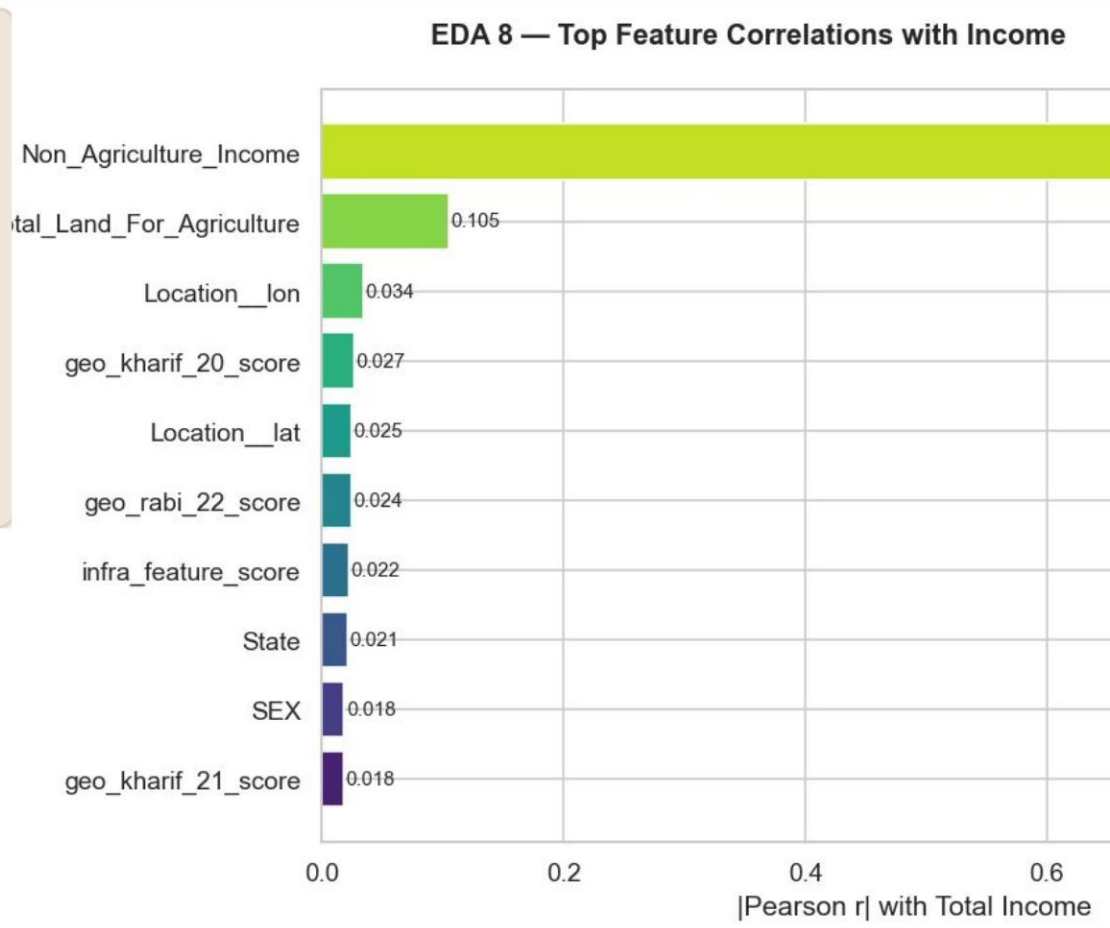
MODELS USED

Random Forest **LightGBM** **XGBoost**

APPROACH

Random Forest provided a strong and stable baseline, while LightGBM and XGBoost, being gradient boosting models, improved accuracy, handled missing values efficiently, and captured feature interactions better — leading to superior performance on the **MAPE metric.**

Model	MAPE %	R²	RMSE (₹)
Random Forest (700 est.)	24.34%	0.3731	1,714,524
LightGBM (mape obj)	20.24%	0.8820	743,916
LightGBM CV (5-fold)	19.61%	–	–
XGBoost (log-MSE)	22.26%	0.7905	991,037
XGBoost CV (5-fold)	21.69%	–	–
Ensemble A: LGB+RF	20.24%	0.8820	743,916
Ensemble B: LGB+XGB+RF	20.42%	0.8776	757,690



BONUS TASK

FROM INSIGHT TO IMPACT

PART A Intervention Recommender

How can we increase income of poor farmers?

FOCUS

Bottom 25% farmers based on predicted income

APPROACH

- Used SHAP to identify key income drivers
- Filtered only controllable variables
- Performed counterfactual simulation (median → upper quartile)
- Recomputed income using trained model

INSIGHT

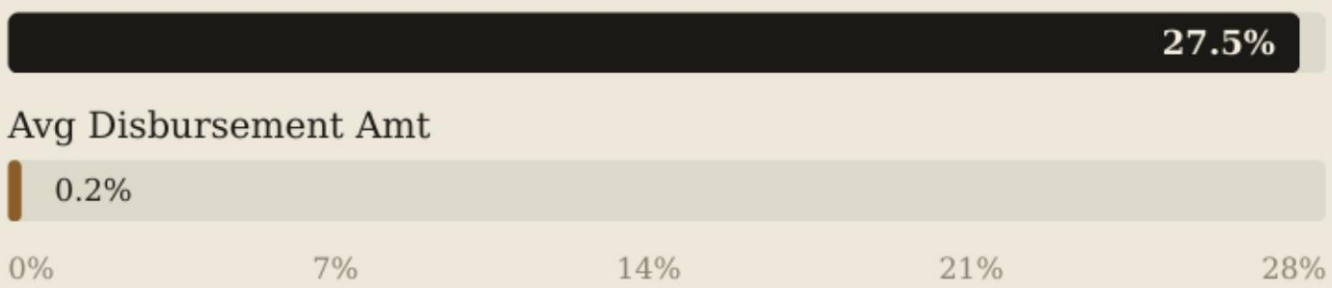
Small, targeted improvements in controllable factors can significantly increase income.

TOP 2 ACTIONABLE INTERVENTIONS

Total Land for Agriculture
Increase usage (median → 75th percentile)
Average Income Uplift
+27.5% ↑

Avg Disbursement Amt
Improve access (median → 75th percentile)
Average Income Uplift
+0.2% ↑

INTERVENTION IMPACT (Average Income Uplift %)



Targeted interventions can uplift income of vulnerable farmers and improve their financial well-being.

PART B Seasonal Volatility Modelling

Which farmers are risky even if income looks okay?

FOCUS

Capture income instability across seasons & years

APPROACH

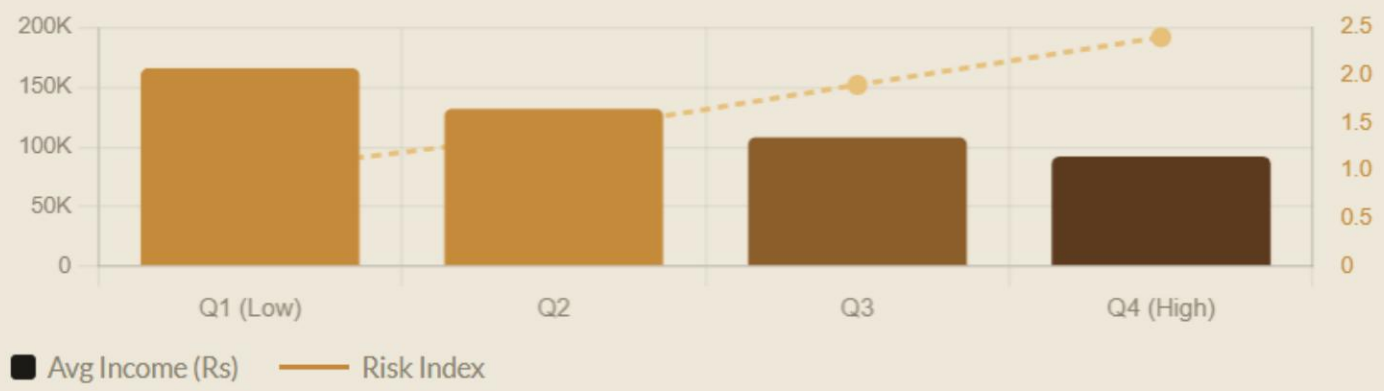
- Converted seasonal data to panel format (farmer-year-season)
- Engineered volatility features
- Created volatility score (normalized)
- Analyzed relationship with income

INSIGHT

Income stability is a stronger indicator of creditworthiness than average income.

KEY FINDINGS

High Volatility → Lower Income
Farmers in highest volatility quartile have **26% lower** average income compared to lowest quartile.

High Volatility → Higher Risk
High volatility farmers are **2.1x more likely** to be in the low-income group.

AVERAGE INCOME BY VOLATILITY QUARTILE